Foundations of Optimization

STAT 525 - Topics in Computational Statistics

Shulei Wang

University of Illinois at Urbana-Champaign

shuleiw@illinois.edu

https://sites.google.com/view/shulei-wang/



Overview

Basics of Optimization Problems

Convex Sets

Convex Functions

Nonsmooth Function and Subgradient

Legendre-Fenchel Conjugate Duality

Optimization Problems

- In the last slides, we show that most statistical and machine learning models can be cast as an optimization problem

$$\min_{x \in \Omega} f(x).$$

- But what does it mean to solve an optimization problem?

Optimization Problems

- When Ω has no restrictions, we call it unconstrained optimization problem

$$\min_x f(x).$$

- We have several definitions of minimizer:
  - Local minimizer: there is a neighborhood N of x^* such that $f(x) \geq f(x^*)$ when $x \in N$.
  - Strict local minimizer: there is a neighborhood N of x^* such that $f(x) > f(x^*)$ when $x \in N$ and $x \neq x^*$.
  - Global minimizer: $f(x) \geq f(x^*)$ for any x.
  - Unique minimizer: x^* is the only global minimizer.

Optimization Problems

- When Ω has some restrictions, we call it constrained optimization problem

$$\min_{x \in \Omega} f(x).$$

- We have several definitions of solution:
  - Local solution: there is a neighborhood N of x^* such that $f(x) \geq f(x^*)$ when $x \in N \cap \Omega$.
  - Global solution: $f(x) \geq f(x^*)$ for any $x \in \Omega$.
- The goal of optimization problems is to search for minimizer or solution.

Taylor's Theorem

- Taylor's theorem is a powerful tool in continuous optimization problem, showing how smooth functions can be approximated locally by polynomials that depend on low-order derivatives of f.
- [**Theorem**] Given a continuously differentiable function f and two points x and p, we have

$$f(x+p) = f(x) + \int_0^1 \nabla f(x+\gamma p)^T p d\gamma,$$
$$f(x+p) = f(x) + \nabla f(x+\gamma p)^T p, \qquad \text{some } \gamma \in (0,1).$$

If f is twice continuously differentiable, we have

$$\nabla f(x+p) = \nabla f(x) + \int_0^1 \nabla^2 f(x+\gamma p) p d\gamma,$$
$$f(x+p) = f(x) + \nabla f(x)^T p + \frac{1}{2} p^T \nabla^2 f(x+\gamma p) p, \qquad \text{some } \gamma \in (0,1).$$

M-Smoothness

- A function f is called M-smooth function or has M-Lipschitz gradients if f is continuously differentiable and

$$\|\nabla f(x) - \nabla f(y)\| \leq M\|x - y\|.$$

- Consider an example $f(x) = x^T Ax$ and f is M-smooth when $\|A\| \leq M/2$.

$$\|2Ax - 2Ay\| \leq \|2A\|\|x - y\|$$

Descent Lemma

- The M-smooth function f can be upper bound by a quadratic function.

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{M}{2}\|y - x\|^2.$$

Proof: By the fundamental theorem of calculus,

$$\begin{aligned}
|f(y) - f(x) - \langle \nabla f(x), y - x \rangle| &= \left| \int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle dt \right| \\
&\leq \int_0^1 \|\nabla f(x + t(y - x)) - \nabla f(x)\| \|y - x\| \rangle dt \\
&\leq \int_0^1 tM\|y - x\|^2 dt \\
&\leq \frac{M}{2}\|y - x\|^2.
\end{aligned}$$

M-Smooth Function

- Suppose f is twice continuously differentiable. Then

$$f \text{ is } M-\text{smooth} \quad \Leftrightarrow \quad \|\nabla^2 f(x)\| \leq M \; (-MI \preceq \nabla^2 f(x) \preceq MI)$$

Proof: If $\|\nabla^2 f(x)\| \leq M$, then

$$\begin{aligned}
\|\nabla f(y) - \nabla f(x)\| &= \left\| \int_0^1 \nabla^2 f(x + t(y-x))(y-x) dt \right\| \\
&\leq \left\| \int_0^1 \nabla^2 f(x + t(y-x)) dt \right\| \|y - x\| \\
&\leq \int_0^1 \left\| \nabla^2 f(x + t(y-x)) \right\| dt \|y - x\| \\
&\leq M \|y - x\|
\end{aligned}$$

M-Smooth Function

Proof: For any given $e \in \mathbb{R}^d$, we have

$$\nabla f(x + \alpha e) - \nabla f(x) = \int_0^\alpha \nabla^2 f(x + te) e dt$$

If f is M-smooth, then

$$\|\nabla f(x + \alpha e) - \nabla f(x)\| \leq M\alpha\|e\|.$$

If we let $\alpha \to 0$, we can show

$$\|\nabla^2 f(x)e\| \leq M\|e\|$$

Since e is arbitrary, we have

$$\sup_{\|e\| \leq 1} \|\nabla^2 f(x)e\| \leq M$$

Characterizing Minima

- In unconstrained optimization problem, what does the minimizer look like?

$$\min_x f(x).$$

- Necessary Conditions for Smooth Unconstrained Optimization
  1. Suppose f is continuously differentiable. If x^* is a local minimizer, then $\nabla f(x^*) = 0$.
  2. Suppose f is twice continuously differentiable. If x^* is a local minimizer, then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive semidefinite.

Proof: We first prove 1. Suppose for contradiction that $\nabla f(x^*) \neq 0$. We can consider another point $x^* - \alpha \nabla f(x^*)$ for some small $\alpha > 0$. By Taylor's theorem, we have

$$\begin{aligned} f(x^* - \alpha \nabla f(x^*)) &= f(x^*) - \alpha \nabla f(x^* - \alpha \nabla f(x^*))^T \nabla f(x^*) \\ &\leq f(x^*) - \frac{\alpha}{2} \|\nabla f(x^*)\|^2 \ (\nabla f \text{ is smooth}) \\ &< f(x^*) \end{aligned}$$

when α is small enough. This contradicts with the fact that $f(x^*)$ is a local minimizer.

Proof: We next prove 2. Suppose for contradiction that $\nabla^2 f(x^*)$ has a negative eigenvalue, so there exists v such that $v^T \nabla^2 f(x^*) v \leq -\lambda$ for some $\lambda > 0$. We can consider another point $x^* + \alpha v$ for some $\alpha > 0$. By Taylor's theorem, we have

$$\begin{aligned} f(x^* + \alpha v) &= f(x^*) - \alpha \nabla f(x^*)^T v + \frac{1}{2}\alpha^2 v^T \nabla^2 f(x^* + \gamma \alpha v) v \\ &\leq f(x^*) - \frac{\alpha^2}{4}\lambda \ (\nabla^2 f \text{ is smooth}) \\ &< f(x^*). \end{aligned}$$

This contradicts with the fact that $f(x^*)$ is a local minimizer.

Characterizing Minima

- Sufficient Conditions for Smooth Unconstrained Optimization
  1. If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite, then x^* is a strict local minimizer.

Proof: The idea is still to apply Taylor's theorem. Because $\nabla^2 f(x^*)$ is positive definite and $\nabla^2 f$ is continuous, there exists ρ and ϵ such that for any p with $\|p\| \leq \rho$, the eigenvalues of $\nabla^2 f(x^* + p)$ are below bounded by ϵ. By Taylor's theorem, for any p with $\|p\| \leq \rho$, we have

$$\begin{aligned} f(x^* + p) &= f(x^*) + \nabla f(x^*)^T p + \frac{1}{2} p^T \nabla^2 f(x^* + \gamma p) p \\ &\geq f(x^*) + \frac{1}{2} \epsilon \|p\|^2 \\ &> f(x^*). \end{aligned}$$

If we choose $\mathcal{N} = \{x^* + p : \|p\| \leq \rho\}$, then we can know $f(x) > f(x^*)$ for all $x \in \mathcal{N}$ and $x \neq x^*$. Therefore, x^* is a strict local minimizer.

Overview

Basics of Optimization Problems

Convex Sets

Convex Functions

Nonsmooth Function and Subgradient

Legendre-Fenchel Conjugate Duality

Convex Optimization

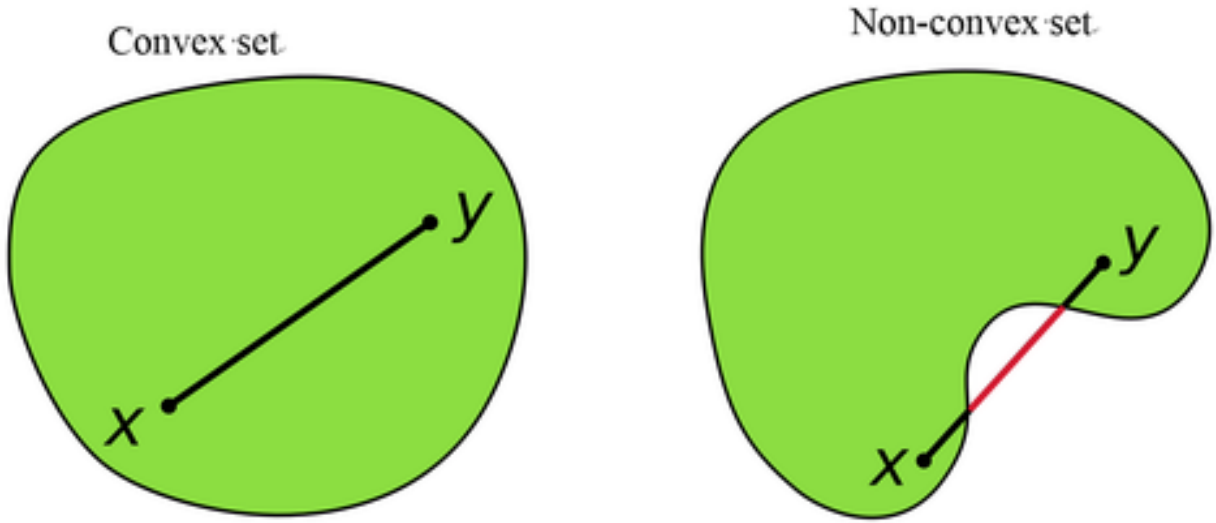
- In general, we cannot guarantee whether we can find an optimal solution.

$$\min_{x \in \Omega} f(x).$$

- We focus a special but broad class optimization problems, convex optimization. In convex optimization, we assume Ω is convex and $f(x)$ is convex (we will explain this later).

Definition of Convex Set

- A set Ω is called convex set if for all $x, y \in \Omega$, then $tx + (1-t)y \in \Omega$, where $0 \leq t \leq 1$.



Some Examples of Convex Set

Here are some examples of convex sets

- d dimentional Euclidean space
- Nonnegative orthant $\mathbb{R}^d_+ = \{x \in \mathbb{R}^d : x_i \geq 0, i = 1, \ldots, d\}$
- Affine subspace, $\{x \in \mathbb{R}^d : Ax = b\}$
- Unit ball, $\{x \in \mathbb{R}^d : \|x\| \leq 1\}$
- Polyhedron, $\{x \in \mathbb{R}^d : Ax \leq b\}$
- ...

Operations Preserving Convexity

Here are some operations that preserves convexity

- Intersection: if Ω_α is convex for all $\alpha \in \mathcal{A}$, then

$$\bigcap_{\alpha \in \mathcal{A}} \Omega_\alpha$$

  is also convex. Actually, any closed convext set Ω is the intersection of all half spaces

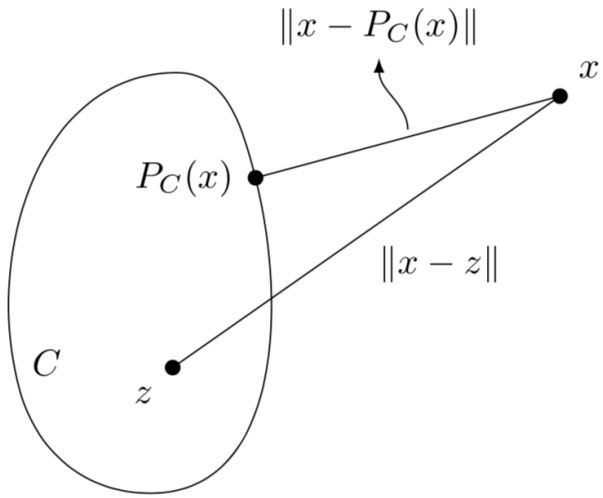
$$\Omega = \bigcap \{H : H \text{ is halfspace, } \Omega \subset H\}.$$

- Affine functions: a function is called affine function if $f(x) = Ax + b$, where A is a matrix and b is a vector. If Ω is convex, then

$$f(\Omega) = \{f(x) : x \in \Omega\} \quad \text{and} \quad f^{-1}(\Omega) = \{x : f(x) \in \Omega\}$$

  are also convex.

Projection onto Convex Set



- Let $\Omega \subset \mathbb{R}^d$ be a closed convex set and $x \in \mathbb{R}^d$ be a point. We then can define the projection of x onto Ω as

$$\text{Proj}_\Omega(x) = \underset{x' \in \Omega}{\text{argmin}} \, \|x - x'\|^2.$$

Projection onto Convex Set

- **[Proposition]** Let $\Omega \subset \mathbb{R}^d$ be a closed convex set and $x \in \mathbb{R}^d$ be a point. Then, $\text{Proj}_\Omega(x)$ must exist.

Proof: For any give x, we define $x_i' \subset \Omega$ as a sequence such that

$$\|x - x_i'\| \to \inf_{x' \in \Omega} \|x - x'\|, \qquad i \to \infty$$

It is not difficult to show that $\{x_i'\}$ is bounded. By Bolzano-Weierstrass theorem, we know there exists a subsequence of $\{x_i'\}$, denoted by $\{\tilde{x}_i'\}$, such that

$$\tilde{x}_i' \to \tilde{x}.$$

As $\{\tilde{x}_i'\} \subset \Omega$ and Ω is closed, we can know that $\tilde{x} \in \Omega$ and

$$\|x - \tilde{x}\| = \lim_i \|x - \tilde{x}_i'\| = \inf_{x' \in \Omega} \|x - x'\|.$$

Projection onto Convex Set

- **[Proposition]** Let $\Omega \subset \mathbb{R}^d$ be a closed convex set and $x \in \mathbb{R}^d$ be a point. Then, $\text{Proj}_\Omega(x)$ is unique.

Proof: Let $d = \inf_{x' \in \Omega} \|x - x'\|$. Suppose there exist two points a and b in Ω such that

$$\|x - a\| = \|x - b\| = d.$$

We define $z = (a + b)/2$ and know $z \in \Omega$ since Ω is convex. The definition of d suggests that $\|x - z\| \geq d$. Observe that

$$\underbrace{\|(x-a)+(x-b)\|^2}_{\|2(x-z)\|^2 \geq 4d^2} + \underbrace{\|(x-a)-(x-b)\|^2}_{\|a-b\|^2} = \underbrace{2\|x-a\|^2 + 2\|x-b\|^2}_{4d^2}.$$

Hence, we have $\|a - b\| \leq 0$ and show $a = b$.

Examples

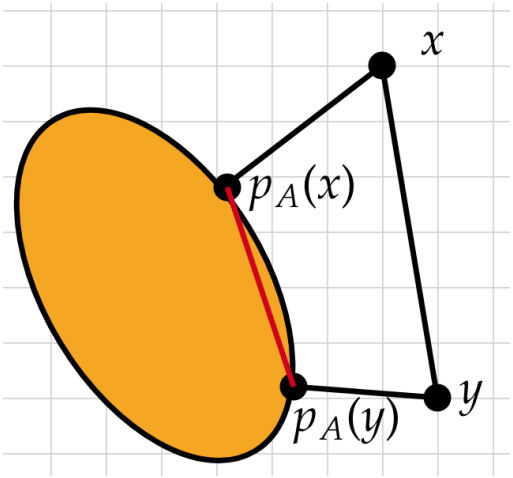
- If $\Omega = \{x \in \mathbb{R}^d : a^T x + b = 0\}$, then

$$\text{Proj}_\Omega(x) = x - \frac{a^T x + b}{\|a\|^2} a.$$

- If $\Omega = \{x \in \mathbb{R}^d : \|x\| \leq 1\}$, then

$$\text{Proj}_\Omega(x) = \frac{1}{\|x\|} x.$$

Properties of Projection



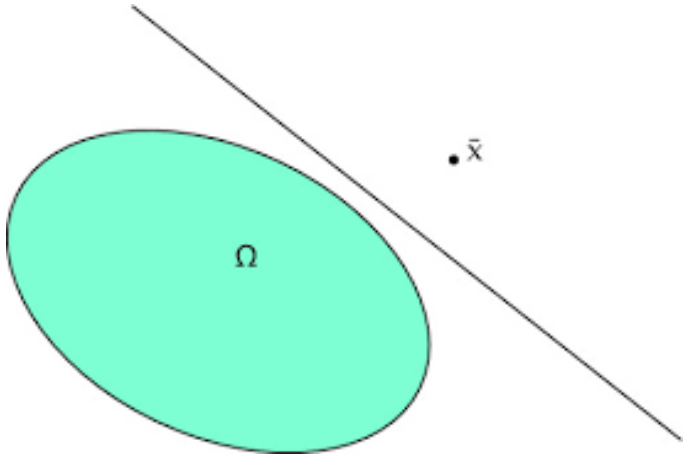
- (firm nonexpansiveness) For any x, y,

$$\langle \mathrm{Proj}_\Omega(x) - \mathrm{Proj}_\Omega(y), x - y \rangle \geq \|\mathrm{Proj}_\Omega(x) - \mathrm{Proj}_\Omega(y)\|^2$$

- (nonexpansiveness) For any x, y,

$$\|\mathrm{Proj}_\Omega(x) - \mathrm{Proj}_\Omega(y)\| \leq \|x - y\|.$$

Separating Theorem



- [**Theorem**]Let $\Omega \subset \mathbb{R}^d$ be a closed convex set and $x \in \mathbb{R}^d$ be a point such that $x \notin \Omega$. Then, there exists $w \in \mathbb{R}^d$ such that

$$\langle w, x' \rangle < \langle w, x \rangle, \qquad x' \in \Omega.$$

If we choose b as a constant between $\sup_{x' \in \Omega} \langle w, x' \rangle$ and $\langle w, x \rangle$, then x and Ω can be separated by the hyperplane $\langle w, y \rangle = b$.

Separating Theorem

Proof: The idea is to show that $w = x - \text{Proj}_\Omega(x)$ can separate x and Ω. Note that $w \neq 0$ because $x \notin \Omega$. For any $x' \in \Omega$, we define $z = tx' + (1-t)\text{Proj}_\Omega(x)$ for any $t \in [0,1]$. Since Ω is convex, we have $z \in \Omega$. By the definition of projection, we have

$$\|x - \text{Proj}_\Omega(x)\|^2 \leq \|x - z\|^2 = \|w - t(x' - \text{Proj}_\Omega(x))\|^2.$$

The right hand side can be seen as a function of t when x and x' are fixed, i.e., $\phi(t) = \|w - t(x' - \text{Proj}_\Omega(x))\|^2$. The above inequality suggests $\phi(0) \leq \phi(t)$ for any $t \in [0,1]$. Because $\phi(t)$ is a smooth function, we must have

$$0 \leq \phi'(0) = -2w^T(x' - \text{Proj}_\Omega(x)).$$

We can obtain the conclusion by reorganizing the above inequality

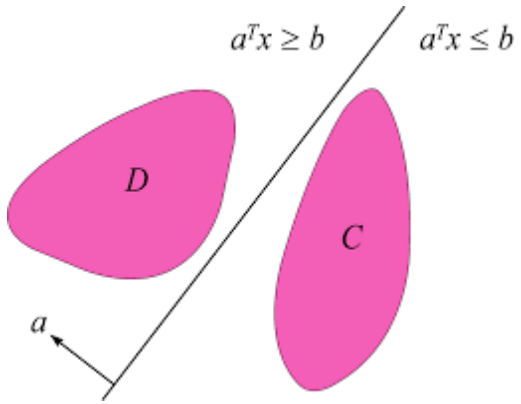
$$w^T x' \leq w^T \text{Proj}_\Omega(x) = w^T(x - w).$$

Separating Theorem

- [**Theorem**]Let $\Omega_1, \Omega_2 \subset \mathbb{R}^d$ be two nonempty closed convex set and $\Omega_1 \cap \Omega_2 = \emptyset$. If Ω_2 is bounded, then, there exists $w \in \mathbb{R}^d$ such that

$$\sup_{x \in \Omega_1} \langle w, x \rangle < \inf_{x \in \Omega_2} \langle w, x \rangle.$$

  If we choose b as a constant between $\sup_{x \in \Omega_1} \langle w, x \rangle$ and $\inf_{x \in \Omega_2} \langle w, x \rangle$, then Ω_1 and Ω_2 can be separated by the hyperplane $\langle w, y \rangle = b$.



Separating Theorem

Proof: Because Ω_1 and Ω_2 are convex,
$\Omega_1 - \Omega_2 = \{x - x' : x \in \Omega_1, x' \in \Omega_2\}$ is also convex. Because Ω_1 is closed and Ω_2 is compact, then $\Omega_1 - \Omega_2$ is closed. Since $\Omega_1 \cap \Omega_2 = \emptyset$, we know that $0 \notin \Omega_1 - \Omega_2$. The separating theorem suggests

$$\sup_{x - x' \in \Omega_1 - \Omega_2} w^T (x - x') \leq 0.$$

Equivalently, we have

$$0 \geq \sup_{x \in \Omega_1} w^T x + \sup_{x' \in \Omega_2} w^T (-x') = \sup_{x \in \Omega_1} w^T x - \inf_{x' \in \Omega_2} w^T (x')$$

We can complete the proof by noting Ω_2 is compact.

Separating Theorem

- [**Theorem**] Let $\Omega_1, \Omega_2 \subset \mathbb{R}^d$ be two nonempty convex set and $Int(\Omega_1) \cap Int(\Omega_2) = \emptyset$. Then

$$\sup_{x \in \Omega_1} \langle w, x \rangle \leq \inf_{x \in \Omega_2} \langle w, x \rangle$$

$$\inf_{x \in \Omega_1} \langle w, x \rangle < \sup_{x \in \Omega_2} \langle w, x \rangle$$

- We omit proof here.

Overview

Basics of Optimization Problems

Convex Sets

Convex Functions

Nonsmooth Function and Subgradient

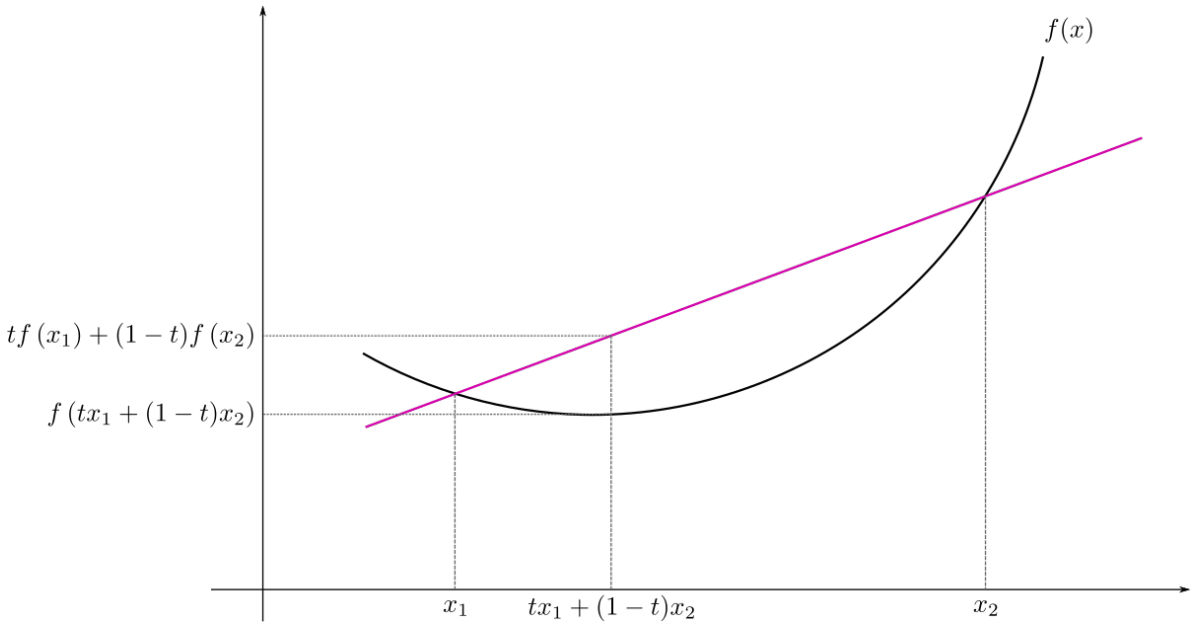
Legendre-Fenchel Conjugate Duality

Definition of Convex Functions

- Let Ω be a given convex set. A function f is said to be convex if for any given $x, x' \in \Omega$ and $0 \leq t \leq 1$, we have

$$f(tx + (1-t)x') \leq tf(x) + (1-t)f(x').$$

- f is said to be strictly convex if we replace $\leq$ with $<$.



Examples of Convex Functions

Here are several exmaples of convex functions

- Exponential function, $f(x) = \exp(ax)$
- Negative logarithm function, $f(x) = -\log x$
- Affine function, $f(x) = \langle w, x \rangle + b$
- Norm function, $f(x) = \|x\|$
- ...

Operations Preserving Convexity

Here are several operations that preserve convexity

- Nonnegative weighted sums: Let f_i be some convex functions and w_i be some nonnegative weights for $i = 1, \ldots, m$. Then

$$f = w_1 f_1 + \ldots + w_m f_m$$

  is convex.

- Affine mapping: If $f(x)$ is convex, then

$$f(Ax + b)$$

  is also convex.

Operations Preserving Convexity

Here are several operations that preserve convexity

- Pointwise maximum: Let f_i be some convex functions for $i = 1, \ldots, m$. Then

$$f(x) = \max\{f_1(x), \ldots, f_m(x)\}$$

  is convex.

- Minimization: If $f(x, y)$ is convex function in (x, y) and Ω' is convex set, then

$$g(x) = \min_{y \in \Omega'} f(x, y)$$

  is convex.

Differentiable Convex Function

- If f is differentiable, then f is (strictly) convex if and only if

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \quad (f(y) > f(x) + \langle \nabla f(x), y - x \rangle).$$

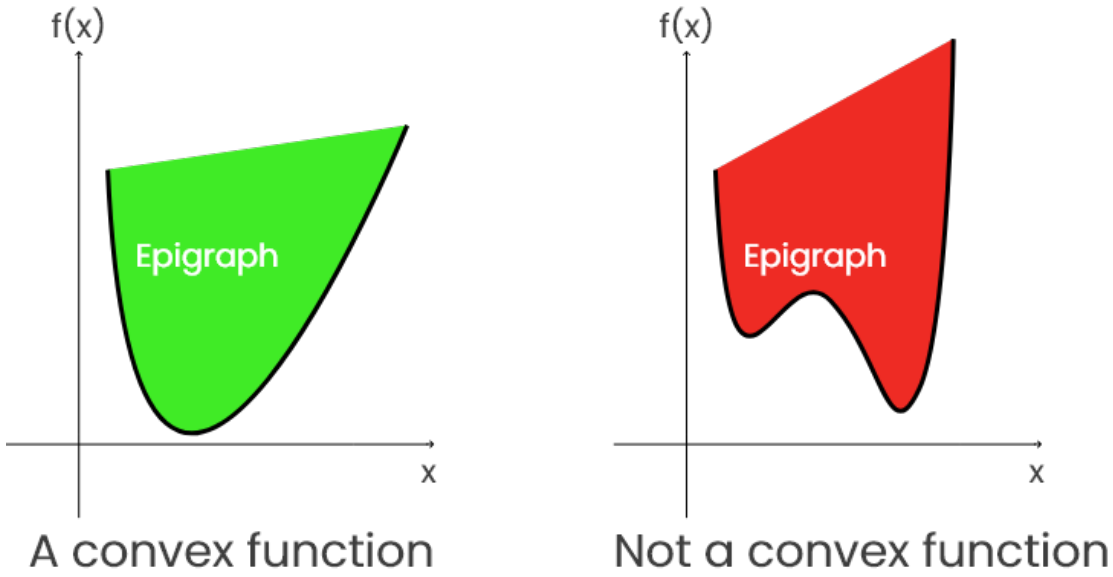
- If f istwice differentiable, then f is (strictly) convex if and only if its Hessian matrix is positive semidefinite

$$\nabla^2 f(x) \succeq 0 \qquad (\nabla^2 f(x) \succ 0).$$

Convex Function

- The epigraph of a function f is given by

$$\text{epi}(f) = \{(x, t) : t \geq f(x)\}.$$



- f is convex if and only if $\text{epi}(f)$ is convex.

Convex Function

- In convex optimization, there is no distinction between "local" and global minimizer/solution.
- [**Theorem**] If f is convex and Ω is closed and convex, then
  1. Any local minimizer is also global minimizer.
  2. The set of global minimizers is convex.

Proof: We first prove 1. Suppose for contradiction that x^* is a local minimizer but there exist $\bar{x}$ such that $f(\bar{x}) < f(x^*)$. Then, the convexity of f suggests that for any $\alpha \in [0, 1]$,

$$f(x^* + \alpha(\bar{x} - x^*)) \leq (1 - \alpha)f(x^*) + \alpha f(\bar{x}) < f(x^*).$$

For any neighborhood $\mathcal{N}$, we can always choose sufficiently small $\alpha > 0$ such that $x^* + \alpha(\bar{x} - x^*) \in \mathcal{N} \cap \Omega$ and $f(x^* + \alpha(\bar{x} - x^*)) < f(x^*)$. This contradicts the definition of local minimizer.

Proof: We next prove 2. Suppose we have two global minimizers x_1^* and x_2^* with $f(x_1^*) = f(x_2^*)$. The convexity of f suggests

$$f((1-\alpha)x_1^* + \alpha x_2^*) \leq (1-\alpha)f(x_1^*) + \alpha f(x_2^*) = f(x_1^*).$$

Because of the definition of global minimizer, we have

$$f((1-\alpha)x_1^* + \alpha x_2^*) \geq f(x_1^*).$$

Therefore, we can conclude

$$f((1-\alpha)x_1^* + \alpha x_2^*) = f(x_1^*).$$

So $(1-\alpha)x_1^* + \alpha x_2^*$ is also a global minimizer.

Convex Function

- It is sufficient to find the local minimizer.
- [**Theorem**] If f is continuously differentiable and convex, then x^* is a global minimizer when $\nabla f(x^*) = 0$.

Proof: For any y, we have

$$f(y) \geq f(x^*) + \langle \nabla f(x^*), y - x^* \rangle = f(x^*).$$

Therefore, x^* is global minimizer.

M-Smooth Function

Suppose f is convex and differentiable. The following conditions are equivalent.

1. f is M-smooth.
2. $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{M}{2}\|y - x\|^2$ for all x, y.
3. $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2M}\|\nabla f(y) - \nabla f(x)\|^2$ for all x, y.
4. $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{1}{M}\|\nabla f(y) - \nabla f(x)\|^2$ for all x, y.
5. $f(tx + (1-t)y) \geq tf(x) + (1-t)f(y) - \frac{M}{2}t(1-t)\|y - x\|^2$ for all x, y.

M-Smooth Function

Proof: "$1 \Rightarrow 2$" can be proved by descent lemma.
"$2 \Rightarrow 3$" Define $v = (\nabla f(y) - \nabla f(x))/M$. By the convexity of f, we have

$$f(y - v) \geq f(x) + \nabla f(x)^T (y - v - x).$$

"2" suggests

$$f(y - v) \leq f(y) + \langle \nabla f(y), -v \rangle + \frac{M}{2} \|v\|^2.$$

Putting above two inequality yields

$$f(x) + \nabla f(x)^T (y - v - x) \leq f(y) + \langle \nabla f(y), -v \rangle + \frac{M}{2} \|v\|^2.$$

We can obtain "3" after rearranging the terms.

Proof: "3 $\Rightarrow$ 4 " "3" suggests

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2M} \|\nabla f(y) - \nabla f(x)\|^2$$

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{1}{2M} \|\nabla f(y) - \nabla f(x)\|^2$$

Adding above two inequalities and rearranging the terms can yield "4".

M-Smooth Function

Proof: "$4 \Rightarrow 1$ " An application of Cauchy -Schwarz inequality on "4" suggests

$$\frac{1}{M}\|\nabla f(y)-\nabla f(x)\|^2 \leq \langle \nabla f(x)-\nabla f(y), x-y\rangle \leq \|\nabla f(x)-\nabla f(y)\|\|x-y\|.$$

rearranging the terms can yield "1".

M-Smooth Function

Proof: "$2 \Rightarrow 5$ " Choose $z = tx + (1-t)y$. We then can apply "2" to obtain

$$f(y) \leq f(z) + t\langle \nabla f(z), y - x \rangle + \frac{Mt^2}{2}\|y - x\|^2$$

$$f(x) \leq f(z) + (1-t)\langle \nabla f(z), x - y \rangle + \frac{M(1-t)^2}{2}\|y - x\|^2$$

Multiplying the first inequality by $1 - t$ and the second inequality by t and adding them can get "5".

Proof: "5 $\Rightarrow$ 2 " Rearranging "5" yield

$$f(y) \leq f(x) + \frac{f(x + (1-t)(y-x)) - f(x)}{1-t} + \frac{M}{2} t \|y - x\|^2.$$

We can complete the proof by taking $t \to 1$.

Strong Convexity

1. f is called m-strongly convex if

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) - \frac{m}{2}t(1-t)\|y-x\|^2.$$

2. Another definition of m-strongly convexity is

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{m}{2}\|y-x\|^2.$$

3. They and the following definition are equivalent.

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq m\|x-y\|^2.$$

Proof: "2 $\Rightarrow$ 1 " Choose $z = tx + (1-t)y$. Then apply "2" to yield

$$f(x) \geq f(z) + (1-t)\langle \nabla f(z), x - y \rangle + \frac{m(1-t)^2}{2}\|y - x\|^2$$

$$f(y) \geq f(z) + t\langle \nabla f(z), y - x \rangle + \frac{mt^2}{2}\|y - x\|^2$$

Multiplying the first inequality by t and the second inequality by $1 - t$ and adding them can get "1".

Proof: "1 $\Rightarrow$ 3 " Rearranging "1" leads to

$$f(y)-f(x)-\frac{m}{2}t\|y-x\|^2 \geq \frac{f(tx+(1-t)y)-f(x)}{1-t} \geq \frac{(1-t)\nabla f(x)^T(y-x)}{1-t}$$

Let $t \to 1$. Then, we have

$$\nabla f(x)^T(y-x) \leq f(y)-f(x)-\frac{m}{2}\|y-x\|^2.$$

With a similar argument, we have

$$\nabla f(y)^T(x-y) \leq f(x)-f(y)-\frac{m}{2}\|y-x\|^2.$$

Adding above two can get "3".

Proof: "3 $\Rightarrow$ 2" Then mean-value theorem suggests

$$f(y) - f(x) = \int_0^1 \nabla f((1-t)x + ty)^T (y - x) dt.$$

"3" leads to

$$\langle \nabla f((1-t)x + ty) - \nabla f(x), y - x \rangle \geq mt\|x - y\|^2.$$

Therefore, we have

$$\begin{aligned} f(y) - f(x) &\geq \int_0^1 \left(\nabla f(x)^T (y - x) + mt\|x - y\|^2 \right) dt \\ &\geq \nabla f(x)^T (y - x) + \frac{m}{2}\|x - y\|^2 \end{aligned}$$

Strong Convexity

- For simplicity, you can roughly consider strongly convex function f as sum of two functions

$$f = g + \frac{m}{2}\|x\|^2,$$

  where g is a convex function.
- We can show f is a strongly convex function if and only if $f - m\|x\|^2/2$ is convex.

*Proof:*Write $g = f - m\|x\|^2/2$. It is sufficient to show f is strongly convex if and only if g is convex, i.e.

$$g(x_t) \leq tg(x) + (1-t)g(y)$$

where $x_t = tx + (1-t)y$. Above inequality is the same with

$$\begin{aligned} f(x_t) &\leq tf(x) + (1-t)f(y) + \frac{m}{2}\left(\|x_t\|^2 - t\|x\|^2 - (1-t)\|y\|^2\right) \\ &= tf(x) + (1-t)f(y) + \frac{m}{2}t(1-t)\|x-y\|^2. \end{aligned}$$

Strong Convexity

Strongly-convex function has some other good properties. Suppose f is m-strongly convex.

- f has a unique minimizer.
- $f(x) - f(x^*) \geq m\|x - x^*\|^2/2$, where x^* is unique minimizer.

We omit proof here. The idea is to show, if x and y are two different minimizer

$$f(y) \geq f(x) + \frac{m}{2}\|y - x\|^2.$$

Overview

Basics of Optimization Problems

Convex Sets

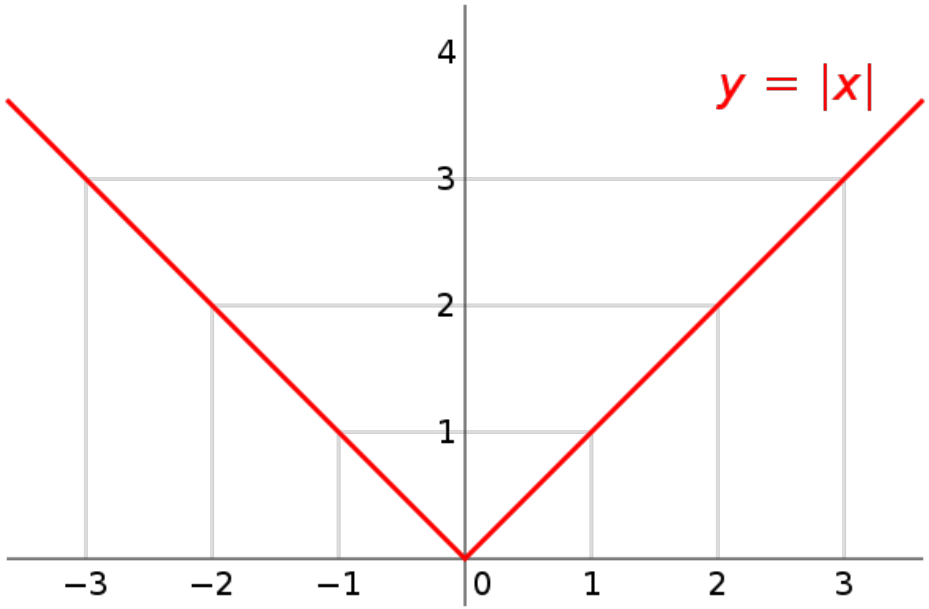
Convex Functions

Nonsmooth Function and Subgradient

Legendre-Fenchel Conjugate Duality

Non-differentialble Convex Function

- f can be convex and non-differentialble, e.g., $f(x) = |x|$.



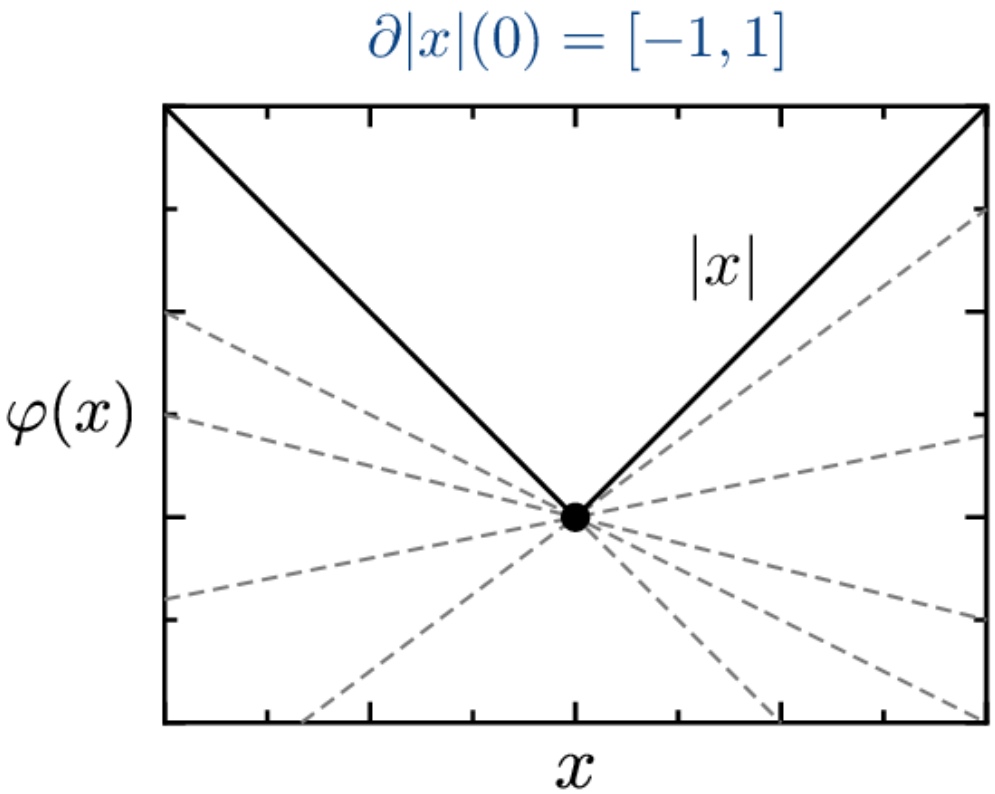
Subgradient and Subdifferential

- When f is non-differentialble, we need to redefine the deravative. Specifically, we introduce an important concept in convex analysis, subgradients.
- g is a subgradient of f at $x \in \Omega$, if for any $y \in \Omega$

$$f(y) \geq f(x) + \langle g, y - x \rangle.$$

- The set of subgradients of f at x is called the subdifferential, which is usually written as $\partial f(x)$.
- When f is differentialble, we can know that $\partial f(x) = \{\nabla f(x)\}$.

Example of Subdifferential



- The subdifferential for $f(x) = |x|$.

Example of Subdifferential

- What is the subdifferential for $f(x) = \|x\|$ at $x = 0$, where $x \in \mathbb{R}^d$?

$$\partial f(x) = \{g : \|g\|_* \leq 1\}, \quad \text{where } \|g\|_* = \max_{\|x\| \leq 1} g^T x.$$

Here, $\|\cdot\|_*$ is called dual norm for arbitrary norm $\|\cdot\|$.

Proof: g is a subgradient of f at $x = 0$, if for any $y \in \Omega$

$$\|y\| \geq \|0\| + \langle g, y \rangle.$$

If $\|g\|_* \leq 1$, then

$$\langle g, y \rangle \leq \|y\| \langle g, y/\|y\| \rangle \leq \|y\| \|g\|_* \leq \|y\|.$$

If $\|y\| \geq \langle g, y \rangle$ for any y, then

$$\|g\|_* = \max_{\|y\| \leq 1} \langle g, y \rangle \leq \max_{\|y\| \leq 1} \|y\| \leq 1.$$

Example of Subdifferential

- Given $p > 1$, the dual norm of l_p is l_q, where $1/p + 1/q = 1$.

Proof: Hölder's inequality suggests

$$\max_{\|x\|_p \le 1} g^T x \le \max_{\|x\|_p \le 1} \|g^T x\|_1 \le \max_{\|x\|_p \le 1} \|g\|_q \|x\|_p \le \|g\|_q.$$

If we take $x_i = \text{sign}(g_i)|g_i|^{q-1}/\|g\|_q^{q/p}$, then we can check

$$\|x\|_p^p = \sum_{i=1}^d (|g_i|^{q-1}/\|g\|_q^{q/p})^p = \sum_{i=1}^d |g_i|^{p(q-1)}/\|g\|_q^q = 1$$

and

$$g^T x = \sum_{i=1}^d g_i \text{sign}(g_i)|g_i|^{q-1}/\|g\|_q^{q/p} = \sum_{i=1}^d |g_i|^q/\|g\|_q^{q/p} = \|g\|_q^{q-q/p}$$

Example of Subdifferential

- The dual norm of l_1 is l_∞ and the dual norm of l_∞ is l_1.

Proof: The proof is similar to the one for l_p and l_q. Note that

$$\max_{\|x\|_1 \leq 1} g^T x = \max_{\sum_i |x_i| \leq 1} \sum_{i=1}^{d} g_i x_i \leq \max_i |g_i|$$

The lower bound can be achieved by setting $x_i = \text{sign}(g_i)$ if $|g_i| = \max_i |g_i|$ and $x_i = 0$ otherwise.

Example of Subdifferential

- Can we find a subgradient for $f(X) = \|X\|$, where $\|\cdot\|$ is the spectral norm and X is a symmetric matrix?

$$vv^T \in \partial f(X)$$

Here, v is the eigenvector of the largest eigenvalue.

Proof: For any symmetric matrix Y,

$$\|Y\| = \max_{\|u\| \le 1} u^T Y u \ge v^T Y v = v^T X v + v^T (Y - X) v = \|X\| + \langle vv^T, Y - X \rangle.$$

Example of Subdifferential

- Subdifferential for spectral norm of matrix $\|X\|$ when $X \in \mathbb{R}^{m \times n}$ for $m \geq n$

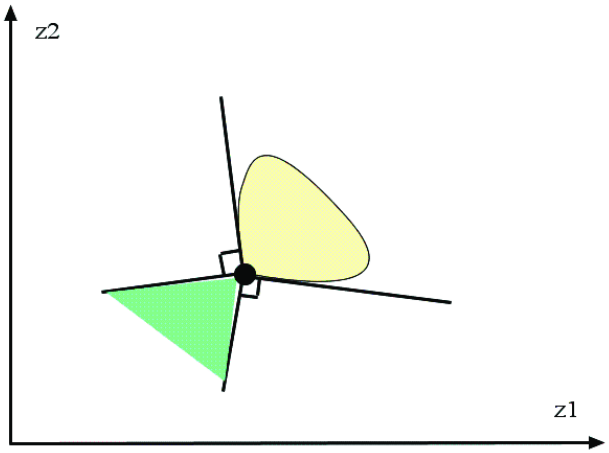
$$\partial \|X\| = \{U_{(1)} H V_{(1)}^T : H \in \mathbb{R}^{t \times t}, H \succeq 0, \mathrm{Tr}(H) = 1\}$$

  where t is the multiplicity of the largest singular value and $U_{(1)}$ and $V_{(1)}$ are the matrix of first t left and right singular vectors.
- Subdifferential for nuclear norm of matrix $\|X\|_*$ when $X \in \mathbb{R}^{m \times n}$ for $m \geq n$

$$\partial \|X\|_* = \{U_{(1)} V_{(1)}^T + U_{(2)} W V_{(2)}^T : W \in \mathbb{R}^{(m-n+s) \times s}, \|W\| \leq 1\}$$

  where s is the multiplicity of zero singular value, $U_{(1)}$ and $V_{(1)}$ are the matrix of first $n - s$ left and right singular vectors and $U_{(2)}$ and $V_{(2)}$ are the matrix of rest left and right singular vectors.
- See Watson, G.A., 1992. Characterization of the subdifferential of some matrix norms. Linear algebra and its applications, 170(0), pp.33-45.

Example of Subdifferential



- Consider the following indicator function

$$I_\Omega(x) = \begin{cases} 0, & x \in \Omega \\ \infty, & x \notin \Omega. \end{cases}$$

- If $x \in \Omega$, the subdifferential of $I_\Omega(x)$ is the normal cone of Ω

$$\partial I_\Omega(x) = \{w \in \mathbb{R}^d : w^T(y - x) \leq 0, \forall y \in \Omega\}.$$

Non-differentialble Convex Function

- [**Proposition**] f is a function defined on a convex set Ω. If $\partial f(x) \neq \emptyset$ for all $x \in \Omega$, then f is convex. If f is convex, then for any $x \in Int(\Omega)$, $\partial f(x) \neq \emptyset$.

Proof: Choose $g \in \partial f(tx + (1-t)x')$ for some t. The definition of subgradient suggests

$$f(x) > f(tx + (1-t)x') + (1-t)g^T(x - x')$$

$$f(x') > f(tx + (1-t)x') + tg^T(x' - x)$$

We can show the convexity of f if we multiply the first one by t and the second one by $1-t$ and find their sum.

Non-differentialble Convex Function

Proof: If f is convex, then epi(f) is convex. The definition of epi(f) suggests $(x, f(x)) \notin Int(\text{epi}(f))$. By the separating hyperplane theorem, we can find $(w, v) \neq 0$ such that

$$w^T x + vf(x) \geq w^T y + vt, \qquad \text{for any} (y, t) \in \text{epi}(f).$$

We can know $v \leq 0$ if we choose (y, t) as $(x, f(x) + 1)$. Because $x \in Int(\Omega)$, we can know $x + \epsilon w \in \Omega$ for arbitrary small enough $\epsilon > 0$. If $v = 0$, then we can choose (y, t) as $(x + \epsilon w, f(x + \epsilon w))$ and obtain

$$w^T x \geq w^T (x + \epsilon w).$$

This suggests $w = 0$ which contradict with $(w, v) \neq 0$. So we can conclude $v < 0$. If we choose (y, t) as $(x', f(x'))$, then we can conclude

$$f(x') \geq f(x) + \frac{1}{v} w^T (x - x').$$

So w/v is subgradient.

Calculus for Subdifferential

- Scaling: $\partial(af) = a\partial f$ for some constant $a > 0$
- Addition: $\partial(f_1 + f_2) = \partial(f_1) + \partial(f_2)$ (under some condition)
- Affine mapping: if $g(x) = f(Ax + b)$, then $\partial g(x) = A^T \partial f(Ax + b)$
- Composition: If $g : \mathbb{R} \to \mathbb{R}$ and $h(x) = g(f(x))$, then $\partial h(x) = g'(f(x))\partial f(x)$.
- Finite pointwise maximum: if $f(x) = \max_{i=1,\ldots,m} f_i(x)$, then

$$\partial f(x) = \text{conv}\{\cup_{i:f_i(x)=f(x)} \partial f_i(x)\}.$$

- General pointwise maximum: if $f(x) = \max_{s \in S} f_s(x)$, then

$$\text{conv}\{\cup_{s:f_s(x)=f(x)} \partial f_s(x)\} \subset \partial f(x).$$

Optimality Conditions for Convex Optimization

- Subdifferential is very useful to characterize minimal point. For convex optimization, we only need to check if $0 \in \partial f(x)$.
- **[Proposition]** f is a convex function. x is a local minimum of f, then x is a global minimum of f. Furthermore, this happens if and only if $0 \in \partial f(x)$.

  Proof: The definition of subgradient suggests

  $$0 \in \partial f(x) \Leftrightarrow f(x') \geq f(x) + 0^T(x' - x), \qquad \forall x'.$$

  If x is a local minimum, we can find a small enough t for any given x',

  $$f(x) \leq f((1-t)x + tx') \leq (1-t)f(x) + tf(x').$$

- This is called Fermat's optimality condition.

Optimality Conditions for Convex Optimization

- We also interesting in the constrained optimization problem

$$\min f(x), \qquad x \in \Omega,$$

  where f is convex and Ω is also convex.
- x^* is the optimal solution if and only if there exists $g^* \in \partial f(x^*)$ such that

$$\langle g^*, y - x^* \rangle \geq 0, \qquad \forall\, y \in \Omega.$$

Proof: It is sufficient to consider the optimization problem

$$\min f(x) + I_\Omega(x).$$

The subdifferential can be written as

$$\partial(f(x) + I_\Omega(x)) = \partial f(x) + \{w \in \mathbb{R}^d : w^T(y - x) \leq 0, \forall y \in \Omega\}.$$

We now complete the proof.

Overview

Basics of Optimization Problems

Convex Sets

Convex Functions

Nonsmooth Function and Subgradient

Legendre-Fenchel Conjugate Duality

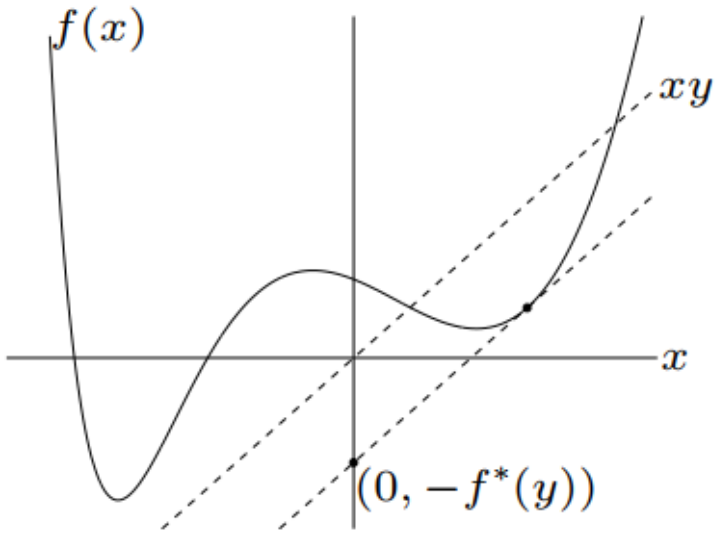
Legendre-Fenchel Conjugate Function

Legendre-Fenchel conjugate function of f is defined as

$$f^*(y) = \sup_x [y^T x - f(x)]$$

f^* is always convex.

$$f^*(y) = \sup_{x \in \mathbf{dom}\, f} (y^T x - f(x))$$



Example of Conjugate Function

Consider an unconstriant quadratic programming problem with a positive definite matrix Q

$$f(x) = \frac{1}{2}x^T Qx.$$

Then the conjugate function is defined as

$$\begin{aligned} f^*(y) &= \sup_x \left[y^T x - \frac{1}{2}x^T Qx \right] \\ &= \frac{1}{2}y^T Q^{-1} y. \end{aligned}$$

See more examples in section 4.4 of First-Order Methods in Optimization.

Properties of Conjugate Function

- We have Fenchel's inequality

$$f(x) + f^*(y) \geq x^T y.$$

- If f is convex and $\text{epi}(f)$ is closed, then

$$(f^*)^* = f.$$

- If f is convex and differentiable, then

$$f^*(y) = {x^*}^T \nabla f(x^*) - f(x^*),$$

where $y = \nabla f(x^*)$.

Properties of Conjugate Function

- Conjugate subgradient theorem: if f is convex and $\text{epi}(f)$ is closed, then

$$y \in \partial f(x) \Leftrightarrow x \in \partial f^*(y) \Leftrightarrow x^T y = f(x) + f^*(y)$$

Proof: If $y \in \partial f(x)$, then

$$f(x') \geq f(x) + y^T(x' - x) \Leftrightarrow y^T x - f(x) \geq y^T x' - f(x')$$

Taking maximum of x' yields

$$y^T x - f(x) \geq f^*(y)$$

By Fenchel's inequality, we show $y^T x - f(x) = f^*(y)$. Then

$$y^T x - f^*(y) = f^{**}(x) \geq x^T y' - f^*(y'), \qquad \forall y'.$$

This suggests $x \in \partial f^*(y)$.

M-Smooth and m-Strongly Convex

The smoothness is also linked to strong convexity in conjugate correspondence.

- If f is $1/m$-smooth convex function, then f^* is m-strongly convex.
- If f is closed and m-strongly convex, then f^* is $1/m$-smooth.